Mikhail Razakov

C++ / LLVM Systems Engineer

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I build systems and compiler components in C++23/C17/Rust: LLVM passes, memory-dependence analysis, CFG/SSA, graph algorithms, x86-64 code generation, and reproducible test oracles

LLVM / C++23
Optimization pass, graph algorithms, CMake, clang-tidy, ASan/UBSan

Conservative analysis
Conservative PS-form memory-dependence analysis with an exact oracle

x86-64 pipeline
SSA-like IR, liveness, linear scan, iced-x86, and differential validation

EXPERIENCE

July 1 — August 31, 2026

MCST — compiler engineering intern

- Implemented an LLVM LICM pass in C++ with conservative checks for side effects, memory access, loop invariance, preheaders, and speculative safety
- Built a Python harness with baseline/optimized execution, opt -verify-each, and explicit CHANGE/NO_CHANGE expectations

2026

PS-form Memory Dependence Analyzer

- Implemented conservative analysis of parametric memory accesses; unresolved cases do not become false yes/no answers, and correctness is checked against an independent exact oracle
- Built an exact oracle for small domains, metamorphic tests, and reproducible WA/TL counterexamples

2026

x86-64 Codegen & Register Allocation Lab

- Designed SSA-like IR, liveness, linear scan, and parity checks between an IR interpreter and generated machine code

PROJECTS

PS-form Memory Dependence Analyzer

Conservative yes/no/maybe results checked by an independent exact oracle on small domains, randomized/metamorphic tests, and reproducible counterexamples.

x86-64 Codegen & Register Allocation Lab

A reference interpreter and differential execution check generated-code semantics; the allocator assignment contract is verified independently.

UniversalToolchain

The modular framework separates language composition from runtime execution and backs its contracts with executable verification/tests.

SKILLS

C++ / toolchain
C++23, C17, CMake, Linux, clang-tidy, ASan/UBSan, GitHub Actions

Compiler analysis
CFG, dominance, SSA, SCC, dataflow, memory dependence, optimization passes

Code generation
x86-64, iced-x86, liveness, linear scan, IR interpretation, isolated execution

Verification
exact oracles, differential/metamorphic tests, stress testing, counterexample reduction

EDUCATION

Software Engineering student at HSE University

RECOGNITION

First-degree diploma and the Grand Prize “Perfection as Hope” for UniversalToolchain.

Moscow / remote